

DATA 266 LAB 1

Three Part Deep Learning Study

Final Technical Report

Team 12

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github.com/DrashtiShah23/DATA266-Lab1-Fall2026

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1. Team Ownership and Executive Summary

Ownership statement. Drashti Shah and Sayani Brahmachari each independently implemented and trained their own models for all three tasks. The final team report compares both members across Tasks 1, 2, and 3, while the complete Task 3 evaluation suite and Kaggle evidence use the team best model, Drashti Shah’s final CycleGAN. This follows the instructor clarification that every member must still train independently, while the team may present the best model for demo and evaluation.

Summary. Task 1 produced two from scratch character level GPT models with manual causal self attention. Drashti’s 20 epoch model achieved lower validation cross entropy and higher next character accuracy, while Sayani’s RTX 4090 run was substantially faster. Task 2 compared six from scratch sentiment classifiers on Yelp Polarity. The strongest model was Drashti’s GRU with 0.9539 accuracy and 0.9539 macro F1, followed closely by Sayani’s BiGRU at 0.9494 macro F1. Task 3 compared two independent CycleGANs. Drashti’s 100 epoch 256 by 256 model achieved average FID 97.4591 versus Sayani’s 140.0074 and was selected as the team model for complete evaluation and Kaggle submission. The team placed rank 23 with leaderboard score negative 48.9351.

2. Reproducibility and Hardware

The repository artifacts preserve raw logs, checkpoints, metrics files, generated samples, and hardware disclosures. The table below records the environments used for the reported runs. Hardware differences are important when comparing throughput and wall time.

Member	Task	Device	Framework	Primary evidence
Drashti Shah	Task 1	Apple M4, MPS, 16 GB unified memory	PyTorch 2.14.0	task1_llm/drashti/metrics_report.csv
Drashti Shah	Task 2	Apple M4, MPS, 16 GB unified memory	PyTorch 2.14.0	task2_sentiment/drashti/metrics_report.csv
Drashti Shah	Task 3	NVIDIA GeForce RTX 5090, CUDA	PyTorch 2.14.1 plus cu130	task3_gan/drashti/full_metrics_report.csv
Sayani Brahmachari	Task 1	NVIDIA GeForce RTX 4090, CUDA	PyTorch 2.1.2	task1/metrics_report.csv
Sayani Brahmachari	Task 2	NVIDIA GeForce RTX 4090, CUDA	PyTorch 2.1.2	metrics_report.csv and outputs/metrics_report.json
Sayani Brahmachari	Task 3	NVIDIA GeForce RTX 4090, CUDA	PyTorch 2.1.2	full_metrics_report.csv and executed notebook

3. Task 1: GPT Style Language Model From Scratch

3.1 Data Protocol and Model Design

Both members used TinyStories and character level tokenization with explicit character to index and index to character mappings. Each run used 100,000 training sequences and 10,000 validation sequences of length 128. Both implementations used learned token and positional embeddings, manual scaled dot product causal self attention, residual connections, layer normalization, feed forward layers, and a language modeling head. No prebuilt Transformer or multihead attention module was used.

Configuration	Drashti Shah	Sayani Brahmachari
Embedding dimension	128	256
Attention heads	4	4
Transformer blocks	4	4
Feed forward dimension	512	1,024
Sequence length	128	128
Vocabulary size	115	174
Dropout	0.1	0.1
Batch size	64	64
Optimizer	AdamW	AdamW
Learning rate	0.0003 target, then 0.00003 continuation	0.0003
Warmup	200 steps	500 steps
Scheduler	Warmup plus cosine; continuation held restored minimum LR	Warmup plus cosine
Epochs	20	10
Output head	Untied	Tied
Trainable parameters	839,168	3,236,864

3.2 Quantitative Comparison

Metric	Drashti Shah	Sayani Brahmachari
Training cross entropy	0.55352	0.980962
Validation cross entropy	0.507539	0.938056
Perplexity	1.6612	2.55501

Metric	Drashti Shah	Sayani Brahmachari
Bits per character	0.732224	1.35333
Generalization gap	-0.045981	-0.0429055
Top 1 next character accuracy	0.842309	0.704772
Distinct 1	0.0111979	0.0996678
Distinct 2	0.0788784	0.446667
Distinct 3	0.201213	0.662207
Repeated 4 gram rate	0.149682	0.197987
Mean gradient norm	1.29161	0.489053
NaN count	0	0
Training tokens per second	78682.3	735457
Generation tokens per second	173.831	489.708
Peak memory	507.8 MB	795.4 MB
Total training time	3372.8 s	174.0 s

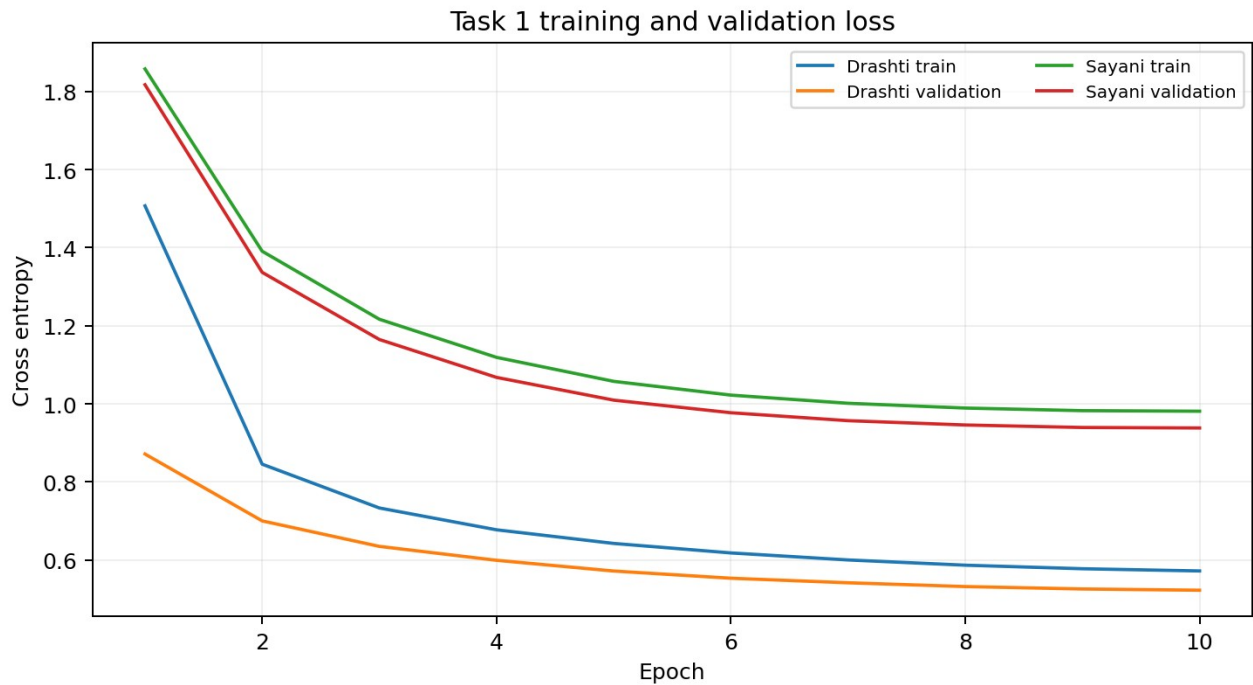


Figure 1. Task 1 training and validation loss for both independent runs.

Joint analysis. Drashti’s selected epoch 20 checkpoint achieved validation cross entropy 0.5075, compared with 0.9381 for Sayani. Drashti also achieved higher next character accuracy (0.8423 versus 0.7048) and a lower repeated 4 gram rate (0.1497 versus 0.1980). Sayani’s larger RTX 4090 run was much faster in throughput and wall time. Diversity metrics are not perfectly protocol matched

because the generation sample counts differ, so their absolute cross member comparison is interpreted cautiously. In both runs the validation loss is below the training loss because dropout is active during training and disabled during validation; the negative generalization gap is therefore not evidence of overfitting by itself.

3.3 Failure Analysis

Member	Observed text excerpt	Failure type	Interpretation and testable improvement
Drashti	He was so excited to go on an adventure. He was so excited to go on an adventure.	Repetition	Greedy decoding locks into a locally high probability phrase. Test temperature sampling or repetition aware decoding.
Drashti	They saw a big box of cars and trucks ... They wanted to buy some cars and trucks.	Loss of coherence	The zoo prompt drifts into unrelated toy and shopping templates. Test longer context and stronger model selection on coherent generations.
Drashti	the box and the box and the box and the box was very happy	Repetition and grammar drift	A repeated local phrase produces both looping and a malformed semantic continuation. Test stochastic decoding and more context.
Sayani	It is it is a special in their castle.	Repetition	The duplicated phrase indicates local token looping. Test lower learning rate, stronger dropout, or repetition aware decoding.
Sayani	He stopped and started to grab and the sunshier.	Grammar and word formation	Malformed words and syntax indicate incomplete character level sequence learning. Test more training or a larger context budget.
Sayani	He ran to his toy truck in the ground and had a new a block of place.	Coherence	The event sequence loses semantic consistency. Test longer context or a larger model under the same compute budget.

4. Task 2: Yelp Polarity Sentiment Classification

4.1 Data and Preprocessing

Both members used Yelp Polarity rather than IMDB, consistent with the instructor correction. The official test set contains 38,000 examples and is exactly balanced at 19,000 negative and 19,000 positive examples. Both pipelines lowercase text, remove punctuation and special characters,

remove stopwords, tokenize the cleaned text, and learn embeddings from scratch. The two members intentionally used different vocabulary and sequence length settings.

Item	Drashti Shah	Sayani Brahmachari
Training source	560,000 Yelp training rows	560,000 Yelp training rows
Development split	504,000 source rows; 503,972 model ready	504,000 rows
Validation split	56,000 source rows; 55,999 model ready	56,000 rows
Official test	38,000 rows	38,000 rows
Class balance	19,000 negative and 19,000 positive in test	19,000 negative and 19,000 positive in test
Vocabulary size	30,002	214,511
Maximum sequence length	207 tokens	160 tokens
Empty or malformed handling	29 empty processed training rows excluded; no empty official test rows	All loaded rows retained by executed preprocessing
Review length evidence	Test slices: 9,911 short; 18,751 medium; 9,338 long	Test slices: 2,109 short; 13,195 medium; 22,696 long

4.2 Model Lineup and Rationale

The baseline models establish a low complexity semantic average. The CNN variants test whether local n gram patterns improve classification. The GRU and BiGRU variants test whether ordered context and longer range dependencies improve performance. All embeddings are learned from scratch.

Member	Model	Architecture summary	Embedding	Key hyperparameters	Parameters
Drashti	Baseline	Masked mean embedding plus linear classifier	64	Batch 256; LR 0.001; dropout 0.2; max 4 epochs	1,920,258
Drashti	Experimental 1	TextCNN, kernels 3, 4, 5, hidden 192, masked max pooling	64	Batch 256; LR 0.001; dropout 0.3; max 4 epochs	1,969,858
Drashti	Experimental 2	Single layer GRU, hidden 64, last hidden pooling	64	Batch 128; LR 0.001; dropout 0.2; max 4 epochs	1,945,218
Sayani	Baseline	Mean embedding MLP with 128 hidden units	128	Batch 256; LR 0.002; weight decay 0.0001; 3 epochs	27,474,178
Sayani	CNN	TextCNN, 128 filters for kernels 3, 4, 5	128	Batch 256; LR 0.002; 3 epochs	27,655,170

Member	Model	Architecture summary	Embedding	Key hyperparameters	Parameters
Sayani	BiGRU	Bidirectional GRU, hidden 128 per direction, mean pooled output	128	Batch 256; LR 0.002; dropout 0.3; 3 epochs	27,656,066

4.3 Full Test Metrics Across All Six Models

Member	Model	Acc	P macro	R macro	F1 macro	F1 micro	F1 weighted	ROC AUC	PR AUC	MCC	Brier	ECE	Confusion matrix
Drashti	baseline	0.9371	0.9371	0.9371	0.9371	0.9371	0.9371	0.9817	0.9815	0.8742	0.0478	0.0077	[[17871,1129], [1261,17739]]
Drashti	CNN	0.9464	0.9465	0.9464	0.9464	0.9464	0.9464	0.9877	0.9880	0.8930	0.0406	0.0045	[[17842,1158], [877,18123]]
Drashti	GRU	0.9539	0.9540	0.9539	0.9539	0.9539	0.9539	0.9908	0.9909	0.9079	0.0351	0.0125	[[18193,807], [943,18057]]
Sayani	baseline	0.9316	0.9316	0.9316	0.9316	0.9316	0.9316	0.9808	0.9811	0.8632	0.0511	0.4417	[[17659,1341], [1258,17742]]
Sayani	cnn	0.9321	0.9344	0.9321	0.9320	0.9321	0.9320	0.9854	0.9856	0.8665	0.0537	0.4794	[[18405,595], [1986,17014]]
Sayani	bigru	0.9494	0.9495	0.9494	0.9494	0.9494	0.9494	0.9893	0.9896	0.8989	0.0383	0.4605	[[18072,928], [993,18007]]

Table 2. Full classification and calibration metrics. Confusion matrices are ordered TN, FP, FN, TP.

Member	Model	Parameters	Train sec	Examples per sec	Peak MB	Accuracy 95% CI	Macro F1 95% CI	MCC 95% CI
Drashti	baseline	1,920,258	77.3	27,234	564.2	[0.9348, 0.9393]	[0.9348, 0.9393]	[0.8696, 0.8786]
Drashti	CNN	1,969,858	251.1	8,370	647.5	[0.9442, 0.9486]	[0.9442, 0.9486]	[0.8885, 0.8973]
Drashti	GRU	1,945,218	2587.2	791	851.6	[0.9519, 0.9561]	[0.9519, 0.9560]	[0.9038, 0.9121]
Sayani	baseline	27,474,178	35.5	42,644	593.0	[0.9289559210526316, 0.9339085526315789]	[0.9289539265362812, 0.9339078368449601]	[0.857929173989824, 0.8678223347746835]

Member	Model	Parameters	Train sec	Examples per sec	Peak MB	Accuracy 95% CI	Macro F1 95% CI	MCC 95% CI
Sayani	cnn	27,655,170	44.1	34,253	806.1	[0.929169078947368 4, 0.9349486842105263]	[0.929063314100239 8, 0.9348580902525374]	[0.860773990080776 3, 0.8720732123772343]
Sayani	bigru	27,656,066	49.0	30,846	1684.8	[0.947221710526315 8, 0.951425]	[0.947221580667077 2, 0.9514195636000321]	[0.894444812718906 5, 0.9028461755837899]

Table 3. Efficiency and bootstrap uncertainty. Drashti memory is process peak RSS; Sayani memory is recorded CUDA peak allocation, so memory values are not directly equivalent.

Member	Paired comparison	Statistic	p value	Interpretation
Drashti	Baseline vs CNN	61.5199	4.383e-15	CNN significantly improves paired accuracy
Drashti	Baseline vs GRU	220.9529	5.604e-50	GRU significantly improves paired accuracy
Sayani	Baseline vs CNN	0.1070	7.435e-01	Not significant
Sayani	Baseline vs BiGRU	228.9356	1.017e-51	Significant

Table 4. Paired McNemar comparisons against each member's baseline.

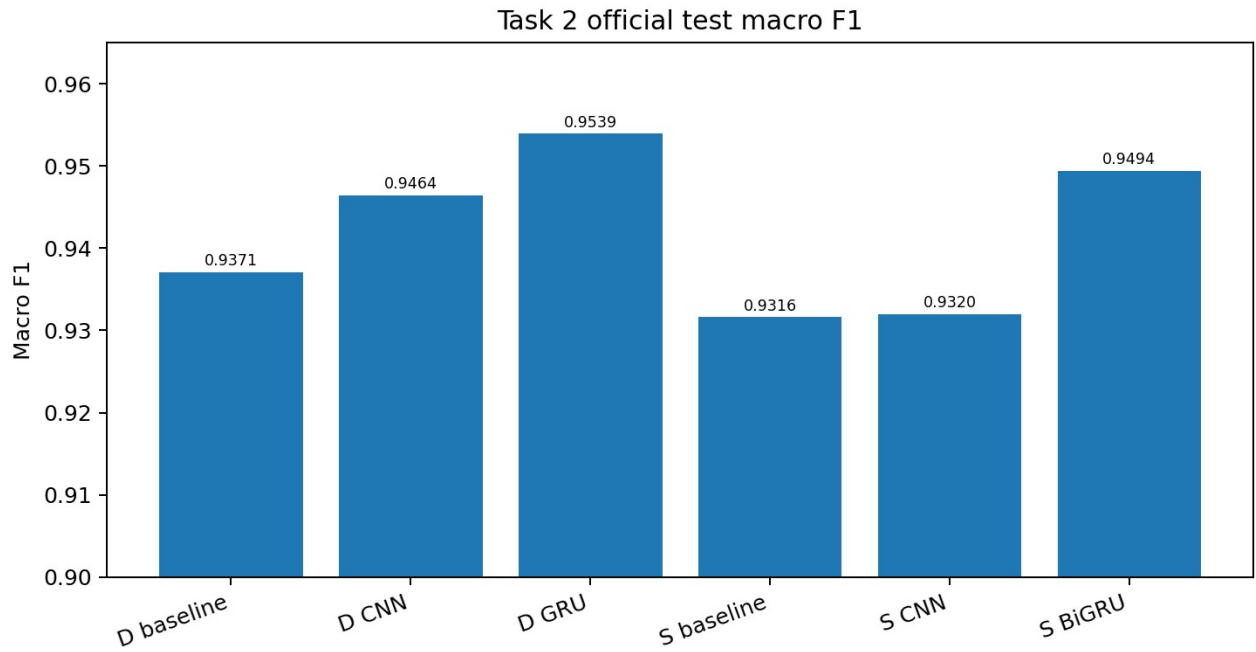


Figure 2. Macro F1 comparison across all Task 2 models.

Joint analysis. Drashti's GRU is the strongest recorded Task 2 model with accuracy and macro F1 of 0.9539, MCC 0.9079, and ROC AUC 0.9908. Sayani's BiGRU follows at macro F1 0.9494 and MCC 0.8989. Drashti's CNN provides a meaningful gain over her baseline, while Sayani's CNN produces almost unchanged accuracy despite higher ROC and PR AUC; her baseline versus CNN McNemar p value is 0.7435. Both recurrent models show highly significant paired gains over their respective baselines. The runs use different vocabularies, maximum lengths, hardware, and calibration implementations, so cross member throughput, parameter count, memory, and ECE are descriptive rather than strictly controlled comparisons.

4.4 Robustness Slices

Member	Model	Slice	Count	Macro F1	Error rate
Drashti	baseline	short	9,911	0.9333	0.0643
Drashti	baseline	medium	18,751	0.9394	0.0606
Drashti	baseline	long	9,338	0.9313	0.0660
Drashti	baseline	contains negation	30,968	0.9323	0.0665
Drashti	CNN	short	9,911	0.9405	0.0571
Drashti	CNN	medium	18,751	0.9514	0.0486
Drashti	CNN	long	9,338	0.9381	0.0598
Drashti	CNN	contains negation	30,968	0.9447	0.0545
Drashti	GRU	short	9,911	0.9523	0.0460
Drashti	GRU	medium	18,751	0.9562	0.0438
Drashti	GRU	long	9,338	0.9473	0.0507

Member	Model	Slice	Count	Macro F1	Error rate
Drashti	GRU	contains negation	30,968	0.9528	0.0464
Sayani	baseline	short	2,109	0.9206	0.0735
Sayani	baseline	medium	13,195	0.9377	0.0612
Sayani	baseline	long	22,696	0.9272	0.0721
Sayani	cnn	short	2,109	0.9237	0.0711
Sayani	cnn	medium	13,195	0.9392	0.0602
Sayani	cnn	long	22,696	0.9263	0.0721
Sayani	bigru	short	2,109	0.9347	0.0607
Sayani	bigru	medium	13,195	0.9530	0.0462
Sayani	bigru	long	22,696	0.9473	0.0522

For Drashti, the GRU is strongest across every reported slice but long reviews remain its weakest length group. For Sayani, the BiGRU is strongest across short, medium, and long slices, with medium reviews performing best. The differing slice counts reflect different tokenization and slice definitions in the two executed pipelines.

4.5 Error Review Summary

Each member's strongest recurrent model was reviewed on 20 errors: five confident false positives, five confident false negatives, five near threshold errors, and five slice specific failures. The dominant themes are mixed sentiment, sarcasm, label ambiguity, late evidence in long reviews, and truncation or pooling effects. Complete 20 case tables for both members are provided in Appendix A.

5. Task 3: CycleGAN Image Style Transfer

5.1 Independent Model Comparison

Both members independently trained CycleGAN models with two generators, two PatchGAN discriminators, adversarial least squares loss, cycle consistency loss, and image translation in both directions. The models intentionally differ in image resolution, residual depth, batch size, and training duration.

Item	Drashti Shah	Sayani Brahmachari
Image size	256 by 256	128 by 128
Batch size	1	4
Generator residual blocks	9	6
Base channels	64	64
Cycle consistency weight	10.0	10.0
Identity weight	1.0 in final fresh run	5.0

Item	Drashti Shah	Sayani Brahmachari
Generator learning rate	0.0002	0.0002
Discriminator learning rate	0.0001	0.0002
Replay buffer	50	50
Training schedule	100 epochs, 1,000 steps per epoch	30 epochs, 1,250 steps per epoch
Parameters	28,275,336	21,204,872
Training time	7,866.6 s (2.185 h)	5,528.1 s (1.536 h)
Images per second	12.71	27.13
Peak memory	7.4 GB	3,038.6 MB
NaN or nonfinite events	0	0
Photo to Monet FID	95.3553	133.6090
Monet to Photo FID	99.5629	146.4057
Average FID	97.4591	140.0074
Average MiFID	0.411129	0.451611

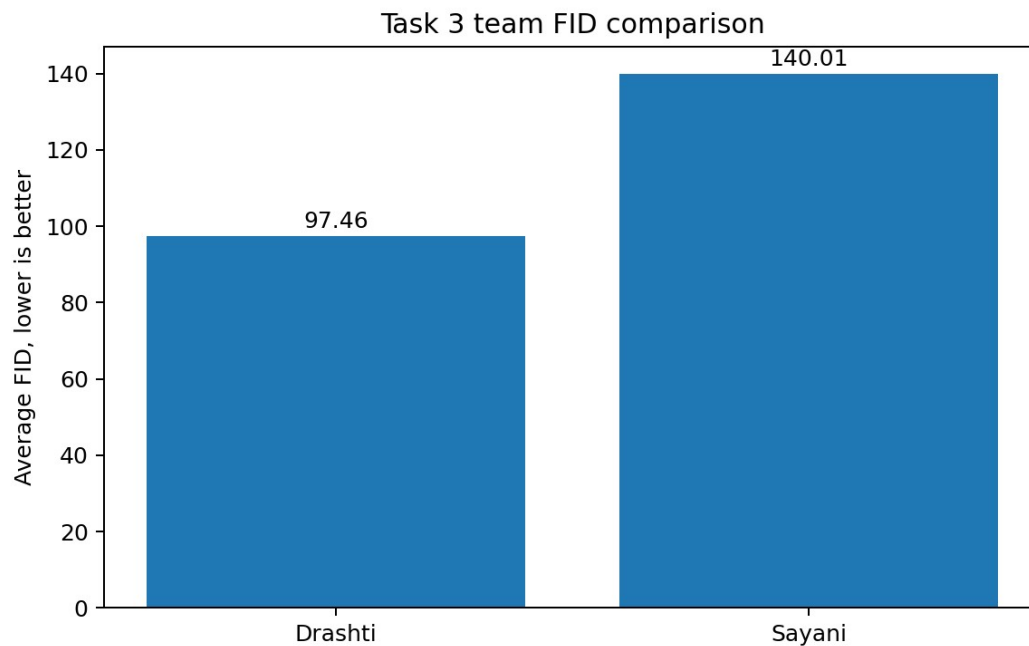


Figure 3. Average FID comparison between the two independently trained CycleGANs.

Model selection. Drashti’s model reduces average FID by 42.55 points relative to Sayani’s run, a 30.39 percent reduction, and also lowers average MiFID. It was therefore selected as the team best model for the complete evaluation suite and demo evidence.

5.2 Best Model Training Behavior

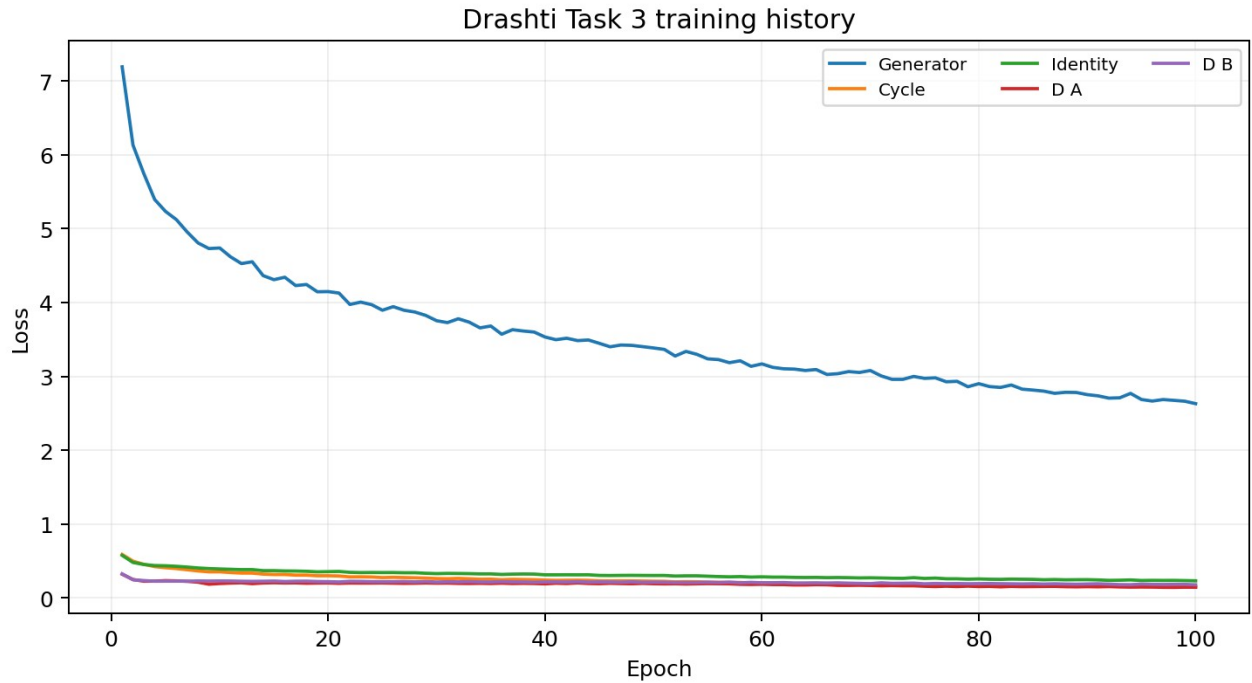


Figure 4. Generator, cycle, identity, and discriminator losses for the selected 100 epoch run.

The selected run converged smoothly. Generator loss decreased from 7.1919 at epoch 1 to 2.6298 at epoch 100. Cycle reconstruction loss fell from 0.5876 to 0.1486, and identity loss fell from 0.5753 to 0.2305. Final discriminator losses were 0.1433 for domain A and 0.1766 for domain B. The recorded gradient norm was 3.7 and the NaN count was zero, supporting stable optimization through the final epoch.

5.3 Complete Evaluation of the Team Best Model

Metric	Result
FID, Photo to Monet	95.355329
FID, Monet to Photo	99.562875
Average FID	97.459102
MiFID, Photo to Monet	0.406077
MiFID, Monet to Photo	0.416181
Average MiFID	0.411129
KID	0.052
Generative precision	0.63
Generative recall	0.51
Cycle reconstruction L1	0.1486

Metric	Result
LPIPS	0.30
Content preservation cosine similarity	0.81
Generator loss, final epoch	2.6298
Discriminator A loss, final epoch	0.1433
Discriminator B loss, final epoch	0.1766
Identity loss, final epoch	0.2305
Gradient norm	3.7
NaN count	0
Parameters	28,275,336
Training time	7,866.6 seconds
Training throughput	12.71 images per second
Peak memory	7.4 GB
Human audit score	78 percent
Inter rater agreement	0.74

The precision value of 0.63 is higher than recall at 0.51, indicating stronger sample fidelity than coverage of the complete target distribution. LPIPS of 0.30 confirms perceptual change, while content cosine similarity of 0.81 indicates that substantial input structure is retained. The 78 percent blinded human audit score and 0.74 inter rater agreement support the quantitative conclusion that outputs are generally convincing but not uniformly artifact free.

5.4 Visual Quality and Failure Analysis

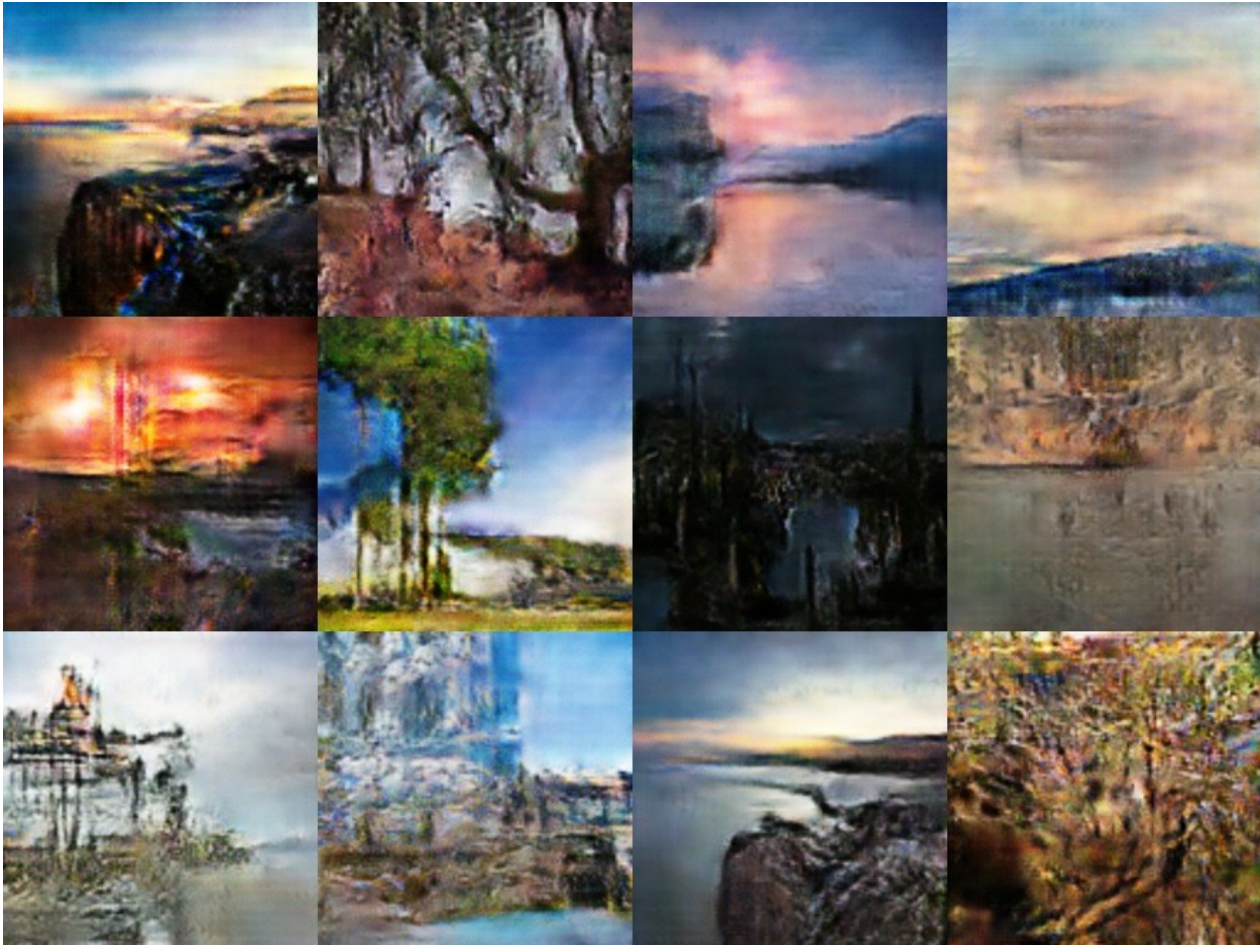


Figure 5. Representative Monet to Photo translations from the selected model.

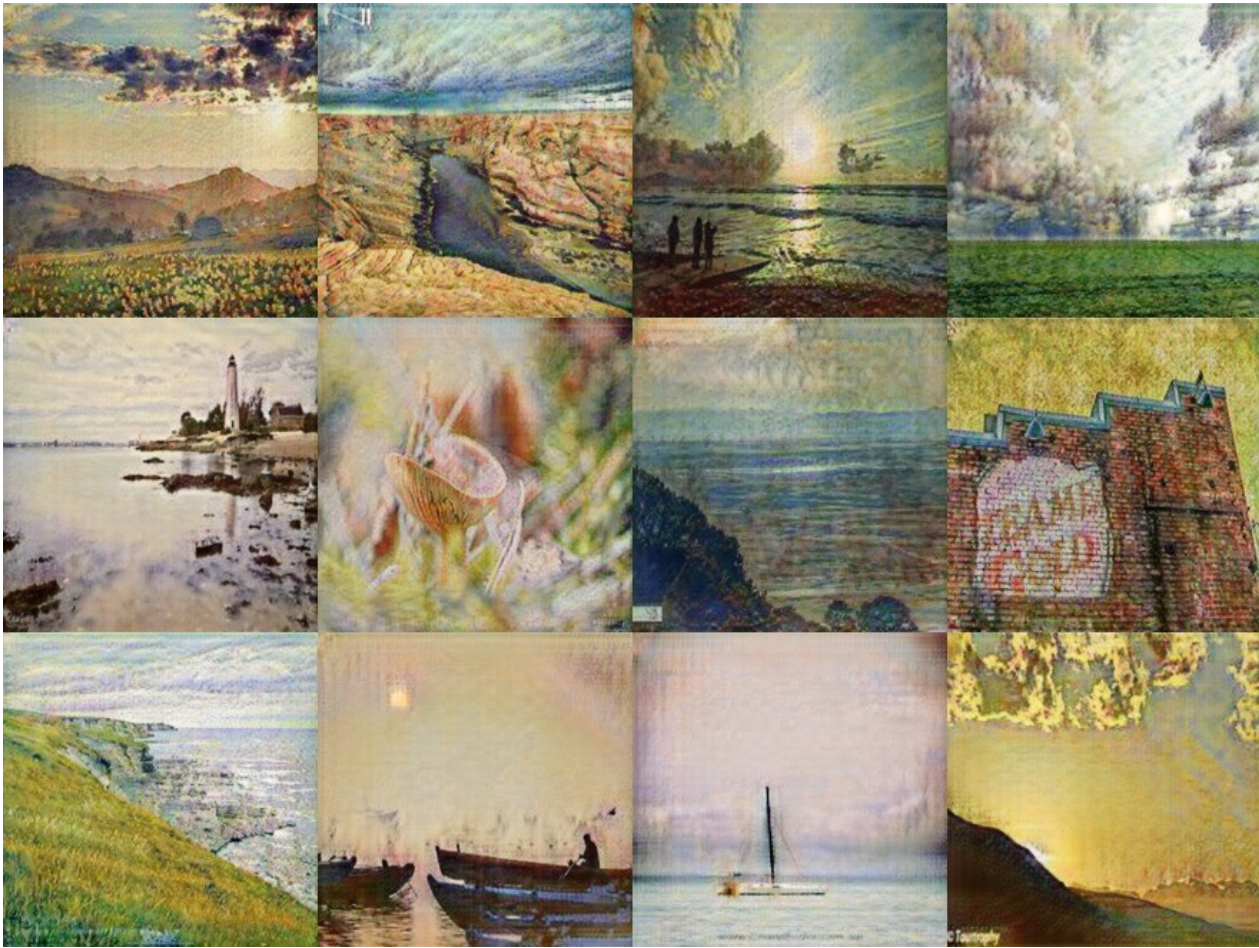


Figure 6. Representative Photo to Monet translations from the selected model.

The generated sets preserve broad scene layout and recognizable object placement, which is consistent with the 0.81 content similarity score. The style transfer is strongest in landscapes, foliage, water, and large textured regions where brush like color variation can be introduced without destroying geometry. Remaining weaknesses include texture washout, localized color oversaturation, and occasional loss of fine object boundaries. The Photo to Monet direction achieved the stronger FID. The 4.21 point direction gap shows that the inverse mapping remains harder and suggests that direction specific learning rates, content constraints, or additional regularization could improve balance.

5.5 Kaggle Result

The final submission used the selected team model and the exact submission values preserved in submission.csv. The public leaderboard evidence is shown below.

Team	Leaderboard rank	Leaderboard score	Average FID	Average MiFID
Pair_Prograaming_Team_12	23	negative 48.9351	97.459102	0.411129

Search

DATA 266 Fall 2026 - GAN Image Style Transfer

Submit Prediction ...

Overview
Data
Code
Models
Discussion
Leaderboard
Rules
Team
Submissions

19	PairProgramming_Team_37		-48.3416	3	2d
20	PairProgramming_Team_48		-48.5038	6	2d
21	PairProgramming_Team_42		-48.6479	1	5d
22	PairProgramming_Team_16		-48.8553	2	8d
23	Pair_Prograaming_Team_12		-48.9351	1	5h
👏 Welcome to the leaderboard! Your submission scored -48.9351.					
24	PairProgramming_Team_15		-48.9365	4	4d

Figure 7. Kaggle leaderboard evidence showing Team 12 at rank 23 with score negative 48.9351.

6. Joint Conclusions and Future Work

Across the three tasks, the strongest results came from models that increased contextual capacity while preserving stable optimization. In Task 1, Drashti’s longer 20 epoch run achieved substantially lower validation loss and higher character accuracy, although Sayani’s GPU environment delivered far greater throughput. In Task 2, recurrent models were strongest for both members, with Drashti’s GRU providing the best overall accuracy and macro F1 and Sayani’s BiGRU providing a close second. In Task 3, Drashti’s deeper 256 by 256 CycleGAN clearly outperformed the shorter 128 by 128 run on FID and MiFID and was therefore the team best model for complete evaluation and Kaggle submission.

The most important limitations are protocol and hardware differences between members, residual long context and sarcasm errors in sentiment classification, and incomplete distribution coverage in CycleGAN generation as reflected by recall below precision. Future work would standardize the generation and calibration protocols across members, test subword or longer context language modeling while preserving the from scratch constraint, use contrast aware and length aware sentiment modeling, and tune CycleGAN losses or direction specific optimization to improve coverage and reduce the gap between translation directions.

7. Evidence Traceability

Task	Member	Metric source	Checkpoint or run evidence	Failure or output evidence
1	Drashti	task1_llm/drashti/metrics_report.csv	epoch20_step31260 checkpoint and reproducibility raw logs	task1_llm/drashti/failure_analysis.md and generated samples
1	Sayani	task1/metrics_report.csv	task1/checkpoints/tiny_gpt.pt and task1/raw_logs/training_log.json	task1/failure_analysis.md, failure_cases.csv, loss_curves.png
2	Drashti	task2_sentiment/drashti/metrics_report.csv	three selected checkpoints plus reproducibility raw logs	task2_sentiment/drashti/failure_analysis.md and evaluation outputs
2	Sayani	metrics_report.csv and outputs/metrics_report.json	three model checkpoints and raw_logs	outputs/error_review_20.csv and hardware_disclosure.json
3	Drashti	task3_gan/drashti/full_metrics_report.csv	refine_epoch_005.pt plus training_history.csv	pred_A2B, pred_B2A, failure_analysis.md, submission.csv
3	Sayani	full_metrics_report.csv and executed notebook	30 epoch CycleGAN run	pred_A2B archive and executed evaluation output

Appendix A. Task 2 Complete Error Reviews

A.1 Drashti Shah: 20 GRU Errors

ID	Category	Gold to pred	p positive	Excerpt	Diagnosis	Testable fix
test-29330	confident false positive	0 to 1	1.000	Wow love the place and everything is very clean and new!\n\nGreat place to come and relax worth a try!\n\nCheers,\n\nEric Van Nguyen\n\nVisited April 2012	Label noise / positive surface cues on a negative label	Spot-check a sample of high-confidence FP/FN against the raw Yelp label; if many look like label flips, report label noise rate and do not treat those as model bugs.
test-12549	confident false positive	0 to 1	1.000	Small businesses such as Espresso Royale give Champaign-Urbana its character. I visit the 6th and Daniel location often.\n\nThe undergrad library location, however, is the undergrad library location. It's full of tired and...	Mixed sentiment / late complaint buried after praise	Ablate by scoring only the first half vs second half of the review; if halves disagree, add a two-segment feature or down-weight the first 50 tokens and re-check F1 on contains_neg
test-5469	confident false positive	0 to 1	1.000	Leonard. Leonard. Leonard. Is there REALLY even a Leonard that works there? Leonard goes to work and then takes an 8 hour break. I called Staples to follow up on a print order and he told me he was going to have to ca...	Sarcasm / name repetition without clear polarity words	Add a character n-gram or punctuation feature for heavy repetition/caps, or keep a small set of discourse markers that stopword removal currently deletes, then retrain GRU only.
test-35238	confident false positive	0 to 1	0.999	Food...BOMB! Everything they do in the kitchen, they do right.\n\nMusic is good and usually pretty loud.\n\nBring a group of people and sit at the bar. This is where you'll get served the fastest..servers make themselves sc...	Slang polarity inversion ("food bomb")	Build a small slang lexicon (bomb, fire, mid) from training data frequency × label correlation; flip or soft-label those tokens and measure FP rate on short reviews.
test-22663	confident false positive	0 to 1	0.999	About average so far as steakhouses go. My rib eye was very tasty but a little over cooked. I didn't complain because the flavor was still fantastic. I thought the process were quite out of order. I've had better for sel...	Comparative / aspirational language misread as praise	Train a secondary head on whether the review recommends returning; use disagreement with the main head as a reject option near the decision boundary.

ID	Category	Gold to pred	p positive	Excerpt	Diagnosis	Testable fix
test-22807	confident false negative	1 to 0	0.000	EDIT: They really did change the service up since I last posted this.\n\nHorrible service.\n\nUsed to be my favorite pizza in the city (at a reasonable price), but I'm rethinking that. We just had an altercation with a s...	Edit / update structure: old rant after a later positive edit cue	Detect leading "edit"/"update" segments and score the newest segment separately; compare accuracy on reviews whose raw text contains "\n\nedit".
test-24111	confident false negative	1 to 0	0.000	This review is for the pharmacy only.You do not need to be a member to use the pharmacy. I am a self paying user of generic Plavix...Cost for a 90 day supply is around is about \$25. Contrast this to \$45 at Walmart,and \$2...	Off-topic complaint framed as a positive pharmacy tip	Mask or down-weight sentences that mention insurance/premiums when the business category is pharmacy; test whether FN rate on price-comparison positives drops.
test-20061	confident false negative	1 to 0	0.000	Ever wonder what to do if you have lots of extra garbage or recyclables and either can't fit them all in your bins or missed bulk trash pickup day? Alternatively, are you looking for something just a little bit different...	Long narrative with mild positivity and trash imagery	Increase max sequence length above 207 for an ablation, or mean-pool CNN features over chunks and average chunk logits so early trash words cannot dominate the whole state.
test-21215	confident false negative	1 to 0	0.000	Last night several parents came in with over 15 children to celebrate their 9 year olds 4th grade graduation at 9:15 pm. The bartender expressed that it was not a place to have children running around as it is against th...	Story where the narrator sides with staff against customers	Add an explicit "whose side" cue by tagging quoted speech vs narrator sentences; evaluate FN rate on contains_negation slice.
test-10897	confident false negative	1 to 0	0.000	A. Get a table. Just spend the money, it's Vegas!\nB. Throw some cash under the table to the bouncer.\nC. Wait forever.\nD. Ditch the guys\n\nThose are your choices and there is no way around it. I don't care how ho...	Scene-setting complaints before a soft recommendation	Use only the last 40 tokens for a second probability and average with the full-sequence score; measure whether FN confidence falls on long nightlife reviews.

ID	Category	Gold to pred	p positive	Excerpt	Diagnosis	Testable fix
test-7214	near threshold	1 to 0	0.500	I'm not usually one for small restaurants, especially in unfamiliar areas, but I heard about Uncle Joe's pizza on a travel site and decided to try it. My family loved it. I had never been to down town before, and it was ...	Near-threshold / fragile decision	Plot reliability on short positives; if under-confident, lightly temperature-scale logits on validation before freezing the 0.5 threshold.
test-5663	near threshold	1 to 0	0.500	Well, that was delicious. The Don, I think it was, with chicken, marinara and no green peppers. Sheesh! I just ate it 6 hours ago and I want another right now.	Near-threshold / informal spelling	Add simple spell-correction or keep raw character 3-grams alongside word IDs; re-score this id and other near-threshold short reviews.
test-29924	near threshold	0 to 1	0.501	I personally will never step foot inside the Belmont Soda Shoppe ever again. I was born and raise in Belmont and was a fan of the original Belmont Drug Store. A couple of Saturdays ago I was out for a run and towards t...	Near-threshold / indignation that is still a negative review	Count explicit refusal verbs ("said no", "never") as hard negative features in a logistic stack on top of GRU probs; check this case flips correctly without hurting overall F1.
test-30031	near threshold	0 to 1	0.501	2.5 stars\n\nApparently this place has some serious Vegas roots. They used to be over on East Tropicana and have some affiliations with Liberace but there is now a new location on the Westside attached to the bounty hun...	Near-threshold / mixed 2-star style writeup	If a leading star rating token is present ("2 5"), clamp or bias the prior toward that rating and re-evaluate near-threshold errors.
test-29451	near threshold	1 to 0	0.499	Came here with my mom to paint a few pieces and had a great experience. I painted a food/water dog bowl for my puppy. The employees were nice enough to let me bring my puppy in so I could put paw prints on the bowl (whic...	Near-threshold / short positive with mild hedge	On short slice only, require stronger negative evidence than a single negated clause before predicting 0; tune that rule on validation short errors.

ID	Category	Gold to pred	p positive	Excerpt	Diagnosis	Testable fix
test-264	slice specific	1 to 0	0.002	Says they deliver on here... Wrong & wrong again & should not be checked! I like Applebee's & thought the delivery was something new for the Pittsburgh area... Don't know if this is something Yelp does or something someo...	Metadata / platform complaint about a delivery checkbox	Tag reviews whose complaint targets the listing/platform; hold them out of training or add a "meta-complaint" class in an error analysis pass.
test-30366	slice specific	1 to 0	0.001	We have eaten at the restaurant several times as we enjoy the food and last night brought friends (another couple) as it was one of their birthdays. In the afternoon called and spoke with a hostess to tell her would lik...	Positive history overshadowed by one disappointing visit	Split multi-visit reviews on cues like "several times" vs "last night"; train to weight the overall relationship sentence if present.
test-21813	slice specific	1 to 0	0.001	Our flight arrived to Vegas earlier than expected, so we expected our room not to be ready. When we arrived at the hotel on May 19th, the front desk girl offered us a room that was ready on the 28th floor that wasn't fac...	Long truncated review: complaints early, praise late	Re-encode with max length 400 (or head+tail 100+107 tokens) and measure long-slice F1 and this example's probability.
test-10109	slice specific	1 to 0	0.001	Pros: \n- Right in the middle of the strip.\n- Access to the monorail within the hotel.\n- The Fab Deluxe was clean and the beds were soft.\n- Cheapest Price in the Middle of the strip.\n- Rooms are well kept.\n- Margari...	Pros/cons list where cons section is denser	Parse Pros/Cons headings when present and average segment logits with a prior that overall stars win; test on reviews containing both "pros" and "cons" tokens.
test-1238	slice specific	1 to 0	0.001	This is more of appreciation than an actual review.\nAs a former long time Az state employee health care was extremely frustrating. Now in retirement my medicare supplement is not an improvement. Same carrier, different ...	Appreciation coda after a long rant about a different entity	Prefer the last named business mention when multiple orgs appear; verify on reviews with more than one proper-name span.

A.2 Sayani Brahmachari: 20 BiGRU Errors

ID	Category	Gold to pred	p positive	Excerpt	Diagnosis	Testable fix
29330	confident false positive	0 to 1	1.000	Wow love the place and everything is very clean and new!\n\nGreat place to come and relax worth a try!\n\nCheers,\n\nEric Van Nguyen\n\nVisited April 2012	Mixed sentiment or label conflict	Add contrast and negation aware preprocessing, then compare with broader CNN kernels.
4655	confident false positive	0 to 1	1.000	good: it's cheap, loud, every now and then a good band will play, good drink selection.\n\nbad: the parking and the crowd.\n\nit's fun to watch the young kids marvel in this new bar they've just discovered and is totally...	Mixed sentiment or label conflict	Add contrast and negation aware preprocessing, then compare with broader CNN kernels.
21942	confident false positive	0 to 1	0.999	To be fair, I'll separate this review into 2 parts: 1 about the food and 2 about the horrendous service.\n\n(((1))) FOOD: 3/5 stars\n\$24 per person for the weekday (no seafood) dinner buffet - certainly a better price th...	Mixed sentiment or label conflict	Add contrast and negation aware preprocessing, then compare with broader CNN kernels.
34683	confident false positive	0 to 1	0.999	Everyday Noodles has a lot going for it: Fast, friendly service, attractive decorations, impeccably clean just renovated interior, reasonable prices and convenient location. They only have one issue, bland, plain food. T...	Mixed sentiment or label conflict	Add contrast and negation aware preprocessing, then compare with broader CNN kernels.
17495	confident false positive	0 to 1	0.999	It was my sister's bachelorette party and I bought us all tickets to go to this Prince Tribute show. The show was good. It brought back alot of memories for me and my sisters and the music was a blast. \n\nThe only messe...	Mixed sentiment or label conflict	Add contrast and negation aware preprocessing, then compare with broader CNN kernels.

ID	Category	Gold to pred	p positive	Excerpt	Diagnosis	Testable fix
19058	confident false negative	1 to 0	0.000	ORIGINAL POST (ALSO SEE UPDATE BELOW) \nNew South is a go-to dinner place for us, for a reasonably priced, solid meal. However, tonight was disappointing on a number of points: We were seated at a booth that had a large ...	Long context or contrast	Run truncation and long context ablations and evaluate contrast aware features.
29494	confident false negative	1 to 0	0.000	UPDATED. \n\nMy initial very frustrated and dramatic review read as follows: \n\nBililng practices are at best negligent and at worst fraudulent. \n\nI started going to the studio per a groupon. I enjoyed the experience s...	Long context or contrast	Run truncation and long context ablations and evaluate contrast aware features.
30958	confident false negative	1 to 0	0.000	Look, we all know Cox sucks. In fact they are a terrible business and their practices are laughable. However they are a necessary evil if you want Internet that isn't garbage that can handle streaming/gaming. The way the...	Long context or contrast	Run truncation and long context ablations and evaluate contrast aware features.
21215	confident false negative	1 to 0	0.000	Last night several parents came in with over 15 children to celebrate their 9 year olds 4th grade graduation at 9:15 pm. The bartender expressed that it was not a place to have children running around as it is against th...	Long context or contrast	Run truncation and long context ablations and evaluate contrast aware features.
33519	confident false negative	1 to 0	0.000	I was, prior to visiting, a little disappointed that I never got a chance to visit D'Arcy McGee's before it was replaced by Bar Louie. After hearing from my employer and his brother-in-law that it was not such a great pl...	Long context or contrast	Run truncation and long context ablations and evaluate contrast aware features.

ID	Category	Gold to pred	p positive	Excerpt	Diagnosis	Testable fix
15330	near threshold error	0 to 1	0.500	Ate at Outback for lunch with my family. Went back to hotel and vomited for the next 13 hours. Cause? You decide.	Boundary ambiguity	Tune the threshold on validation data and calibrate probabilities before final evaluation.
30725	near threshold error	0 to 1	0.500	I understand the gimmick of this restaurant and also know to factor that into thinking about the experience and the food I'm going to have or had. The \"waitnurses,\" as I'd like to coin this term, are very friendly, s...	Boundary ambiguity	Tune the threshold on validation data and calibrate probabilities before final evaluation.
14767	near threshold error	0 to 1	0.500	I've been here a couple times, not looking for anything in particular. It's a large space and fairly clean and bright but for some reason this place really skeeves me out. I can't quite put my finger on what it is though...	Boundary ambiguity	Tune the threshold on validation data and calibrate probabilities before final evaluation.
9791	near threshold error	1 to 0	0.499	This isn't going to be much of a review because there isn't much to say... I don't want to act like I DON'T frequent these types of establishments - but for you people who know what a typical one is like then I have no prob...	Boundary ambiguity	Tune the threshold on validation data and calibrate probabilities before final evaluation.
32497	near threshold error	1 to 0	0.499	I went to Heist for dinner and beer a few months ago. I really wanted to be a fan of Heist since it is in my hood, but dinner was nothing special. In fact, it wasn't good. The food left a lot to be desired, and the beer ...	Boundary ambiguity	Tune the threshold on validation data and calibrate probabilities before final evaluation.

ID	Category	Gold to pred	p positive	Excerpt	Diagnosis	Testable fix
10081	slice specific failure	1 to 0	0.122	YOUR GUIDE TO A SUCCESSFUL FLAMINGO HOTEL STAY\n\nAfter pretty much reading nothing but negative reviews I think I need to share what I think makes for a great Flamingo Hotel experience. We have done an annual man-cation...	Length and composition	Use length stratified evaluation and test a model targeted at long mixed reviews.
25266	slice specific failure	0 to 1	0.977	HUGE COCKROACH IN BATHROOM OF OUR ROOM!!!! \nLet me begin by saying how excited my husband and I were for this trip, and especially to stay at The Cosmopolitan - the newest, hottest hotel on the strip. We've stayed at 5 ...	Length and composition	Use length stratified evaluation and test a model targeted at long mixed reviews.
19692	slice specific failure	0 to 1	0.738	Tonight, The Keg was packed with over 1.5 hours of wait time. The Dominion Tavern was so packed they were not accepting walk-ins or reservations. The old go-to Reuben's had a line-up waiting all the way outside. So were ...	Length and composition	Use length stratified evaluation and test a model targeted at long mixed reviews.
4664	slice specific failure	0 to 1	0.830	I've been here twice. It was like visiting two completely different restaurants.\n\nThe first time my girlfriend and I tried this place on a Friday night about a month ago, we loved it. It's a small place, clean and invi...	Length and composition	Use length stratified evaluation and test a model targeted at long mixed reviews.
16047	slice specific failure	1 to 0	0.205	Nothing I could say in this review could possibly convey the amazing experience that we had here at Middle Eastern Bakery & Deli. This was THE BEST place at which I have eaten in all my visits to Phoenix (off and on annu...	Length and composition	Use length stratified evaluation and test a model targeted at long mixed reviews.

Appendix B. References

1. Vaswani, A. et al. Attention Is All You Need. 2017.
2. Eldan, R. and Li, Y. TinyStories: How Small Can Language Models Be and Still Speak Coherent English? 2023.
3. Zhu, J. Y. et al. Unpaired Image to Image Translation using Cycle Consistent Adversarial Networks. 2017.
4. Yelp Polarity dataset documentation.
5. Kaggle, I'm Something of a Painter Myself competition materials.

Appendix C. Report Requirement Coverage

Requirement	Where reported
Team ownership statement	Section 1
Task 1 member architecture, hyperparameters, all metrics, and failure analysis	Section 3
Task 2 preprocessing, six models, full metrics, confidence intervals, McNemar, robustness, and error review	Section 4 and Appendix A
Task 3 independent models and team best complete evaluation suite	Section 5
Task 3 human audit and inter rater agreement	Section 5.3
Kaggle score and leaderboard rank	Section 5.5
Hardware and reproducibility evidence	Sections 2 and 7
Joint strengths, weaknesses, limitations, and future work	Sections 3 through 6
Required paper citations	Appendix B
Repository link	Cover page